

# What APEX Can Do

A student-facing view of the Semester Planning Agent

Live demo: <https://technion-semester-planner-agent.vercel.app>

APEX is built for the messy part of semester planning: a student has passed some courses, failed others, wants a reasonable workload, may have blocked days, and still needs to make real degree progress. The agent turns that situation into a checked plan and keeps helping when the student changes their mind.

## From One Message To A Plan

The student can write naturally in Hebrew or English. APEX extracts the semester, track, pace, passed courses, failed courses, unavailable weekdays, requested courses, grades, and any special constraints. If something important is missing, it asks for that missing detail instead of guessing.

Once it has enough information, APEX builds a plan with course numbers, course names, credits, exam dates, workload, and a short explanation of the tradeoffs. The final plan is checked by deterministic code before it reaches the student.

## What The Student Gets

| Output               | Why it matters                                                                                                                    |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Verified course list | Courses are checked for prerequisites, duplicates, already-passed courses, unknown numbers, credit range, and mandatory progress. |
| Weekly timetable     | Sections are checked so the plan does not hide class-time collisions.                                                             |
| Exam view            | Moed A and Moed B dates are considered, with warnings when exams are too close.                                                   |
| Workload and risk    | The plan weighs difficulty, review data, heavy-course stacking, and future impact.                                                |
| Human explanation    | The agent explains why this combination was chosen and what, if anything, is still unresolved.                                    |

## Example: A Verified Plan

In this example, APEX produced a complete 18.5-credit semester with a moderate workload score. The student can immediately see the courses, credits, exams, and verification status.

# Semester Plan

הנדסת מערכות מידע

✓ Verified · 18.5 credits · workload 46.8/100

00940224 · 4 pts · exam 2027-02-23

מבני נתונים ואלגוריתמים

00940241 · 3 pts · exam 2027-02-03

ניהול מסדי נתונים

00940424 · 3.5 pts · exam 2027-02-07

סטטיסטיקה 1

00940594 · 3.5 pts · exam 2027-02-17

עקרונות הכלכלה להנדסים

03240521 · 2 pts

יזמות בארגונים-התפתחויות ומגמות

03240675 · 1.5 pts · exam 2027-01-29

ספרדית למתחילים

03940820 · 1 pts

חינוך גופני - תנועה ומחול

Built by my Semester Planning Agent 🤖

A checked semester plan: verified, 18.5 credits, workload 46.8/100.

## Course Advice, Not Just Scheduling

APEX can also answer direct questions about courses. For example, if the student asks whether a course is worth retaking to improve a grade, the answer is based on several signals together: the student's grade, the course's historical average, the student's own average, CheeseFork difficulty, satisfaction, and review text.

This matters because the same grade can mean different things for different students. A 70 can be a warning sign for a student whose average is much higher, but less important for another student whose overall average is close to 70.

you

האם אתה ממליץ לי לשפר קורס? לפי הציונים שלי

AGENT

לפי הנתונים, **סטטיסטיקה 1** הוא מועמד סביר לשיפור: קיבלת בו **70**, שזה קצת מתחת למוצע ההיסטורי של **71.8**, אבל הרבה מתחת למוצע האישי הידוע שלך (**86.5**), כך שהציון הזה בולט לרעה ביחס אליך. מצד שני, המשוב על הקורס חזק מאוד — יש **41** חוות דעת, עם שביעות רצון ממוצעת **4.07/5** וקושי **3.49/5**, וביקורות כמו "**מרצה מדהימה באמת**" ו-"**הכי טובה שיש!**" מרמזות שהקורס עצמו סביר ואף אהוד, כך ששיפור יכול להיות כדאי בעיקר אם אתה חושב שתוכל לחזור קרוב יותר לרמתך הרגילה.

backend: <https://technion-semester-palnn>

The agent explains why Statistics 1 may be worth retaking, using grades, averages, and reviews.

## Follow-Up Changes

APEX does not treat every follow-up as a brand-new planning problem. If the student asks to add, remove, or swap a course, it keeps as much of the previous plan as possible and changes only what needs to change.

When a request creates a real conflict, the agent does not silently drop the request. It explains the conflict and asks for explicit confirmation before forcing an exception.



The screenshot shows a chat window with a light yellow background. At the top, there are two tabs. The main text area contains the following message: "Here's your updated plan. I kept 7 of your 8 courses, removed סדנה בסטטיסטיקה לדוקטורנטים, and added 19.5 — סדרות עתיות וחיווי credits total. Still open: class time overlaps with (11:30-13:30) התנהגות ארגונית in every section combination; only 1 mandatory course(s) in the plan, expected at least 3 - a semester should make real degree progress unless the student asked for fewer. Note: סדרות עתיות וחיווי was removed from the plan because its class times collide with another course in every available section combination - a clashing week is never delivered. ניחול שפה טבעית: you asked to add it, but I couldn't - its class times collide with פרויקטים in every available section combination. Want me to add it anyway despite that? Confirm and I will, no matter what." Below this message is a dashed line and a prompt: "Anything you'd like to change? Pick a quick action or reply below." On the right side of the chat window, there are two search icons and a "backend:" label at the bottom.

Here's your updated plan. I kept 7 of your 8 courses, removed סדנה בסטטיסטיקה לדוקטורנטים, and added 19.5 — סדרות עתיות וחיווי credits total.

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Anything you'd like to change? Pick a quick action or reply below.

A follow-up revision that preserves most of the old plan and clearly names the remaining conflict.

## Retakes And Grades

Failed retakes are treated as high priority because they often block future courses. Grade-improvement retakes are handled more carefully: APEX can suggest one, but it does not add it silently. The student must approve it first.

Grades can come from the chat or from an uploaded transcript PDF. The system does not fetch private Technion account data.

## What Else It Handles

- Three supported Technion tracks.
- Hebrew and English intake.
- Passed, failed, and explicitly requested courses.
- Blocked weekdays and section-level timetable constraints.
- Light, normal, and fast pace preferences.
- Course reviews from CheeseFork.
- Historical grade averages from technion-histograms.
- Transcript PDF parsing into courses and grades.
- Optional Supabase session restore using a hashed student email.
- Live offering check before final delivery.
- Roadmap, risk report, weekly schedule, and exam study plan tools.
- Required API endpoints with prompt, response, and model-step trace.

## Limits And Future Work

APEX is a planning assistant, not a registration system. It does not enroll students, access private Technion accounts, or promise that cached catalog data will update by itself. When Technion publishes

a new catalog, the offline pipeline should be rerun.

Future work could add focused retrieval for academic regulations, forms, and administrative guidance. The core course checks should remain deterministic because prerequisites, schedules, credits, and exam dates need exact answers.